C1.7 Protocol errors

The protocol defines two categories of protocol errors, a software protocol error and a hardware protocol error.

C1.7.1 Software protocol error

A software protocol error occurs when multiple accesses to the same location are made with mismatched shareability or cacheability attributes.

A software protocol error can cause a loss of coherency and result in the corruption of data values. The protocol requires that the system does not deadlock for a software protocol error, and that transactions always progress through a system.

A software protocol error for an access in one 4KB memory region must not cause data corruption in a different 4KB memory region.

For locations held in Normal memory, the use of appropriate barriers and software cache maintenance can be used to return memory locations to a defined state.

When accessing a peripheral device, if Modifiable transactions are used as indicated by **AxCACHE[1]** = 1, then the correct operation of the peripheral cannot be guaranteed. The only requirement is that the peripheral continues to respond to transactions in a protocol compliant manner. The sequence of events that might be needed to return a peripheral device, that has been accessed incorrectly, to a known working state is IMPLEMENTATION DEFINED.

C1.7.2 Hardware protocol error

A hardware protocol error is defined as any protocol error that is not a software protocol error. No support is required for hardware protocol errors.

If a hardware protocol error occurs then recovery from the error is not guaranteed. The system might crash, lock up or suffer some other non-recoverable failure.

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Chapter C2
Signal Descriptions

This chapter introduces the additional ACE interface signals. It contains the following sections:

- *Changes to existing AXI4 channels* on page C2-152
- *Additional channels defined by ACE* on page C2-153
- *Additional response signals and signaling requirements defined by ACE* on page C2-155

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Later chapters define the signal parameters and usage.

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